

Introduction

Multiple imputation (MI; Rubin 1987) replaces each missing value with $m > 1$ plausible values drawn from the posterior predictive distribution of the missing data given the observed. The analysis is run on each of the m completed datasets; Rubin's rules combine the results into a single inference reflecting both within-imputation and between-imputation uncertainty.

Prerequisites

Missing-data mechanisms (MAR); Bayesian predictive distributions.

Theory

MI procedure: 1. Impute: create m completed datasets via a predictive model. 2. Analyse: apply the intended model to each dataset. 3. Pool: combine $\hat{\beta} = (1/m) \sum \hat{\beta}_k$; total variance $T = \bar{W} + (1 + 1/m)B$, where $\bar{W}$ is mean within-imputation variance and B is between-imputation variance.

mice uses chained equations: iteratively impute each variable using the others as predictors.

Assumptions

Missing at random (MAR); imputation model is correctly specified (correct functional form, includes all predictors that might correlate with missingness).

R Implementation

```
library(mice)

set.seed(2026)
n <- 300
df <- data.frame(
  x1 = rnorm(n), x2 = rnorm(n),
  x3 = sample(c("a", "b", "c"), n, replace = TRUE),
  y = rnorm(n)
)
df$y[sample(n, 60)] <- NA      # 20% missing
df$x2[sample(n, 30)] <- NA    # 10% missing

# Chained-equations imputation, 20 imputations
imp <- mice(df, m = 20, method = c("pmm", "pmm", "polyreg", "pmm"),
  printFlag = FALSE)

# Fit the model on each imputation and pool
fit <- with(imp, lm(y ~ x1 + x2 + x3))
pooled <- pool(fit)
summary(pooled)
```

Output & Results

Pooled regression coefficients with SEs that correctly reflect imputation uncertainty; fmi (fraction of missing information) indicates how much of the variance comes from imputation.

Interpretation

“Multiple imputation ($m = 20$) under MAR gave a pooled coefficient of 0.94 (SE 0.10, 95 % CI 0.74-1.14); fraction of missing information 0.18 suggests reasonable efficiency.”

Practical Tips

- Include **all** variables used in the substantive model, plus auxiliary variables correlated with missingness, in the imputation model.
- m should be ≥ 100 when a large fraction is missing; 20 is a minimum for exploratory work.
- For regression on interactions or non-linearity, include those terms in the imputation model too (impute so the model matches).
- Use predictive mean matching (pmm) for continuous variables to avoid unrealistic extrapolations.
- Always check convergence of chained equations via trace plots.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale